

WP2 – Concept, Requirements and Specification

D2.3: User Stories and Requirement Analysis

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Delivery Date: 02/2014 and 01/2015 amended version after MTR

Dissemination Level: Public

Version 1.5

Short Abstract

The main objective of this deliverable is to define the functional requirements for the ALFRED system. This will be done by analysing the end-user requirements gathered from related literature and studies as well as from the ALFRED focus group sessions. Furthermore, the report includes the verified and improved ALFRED use cases and personas that are essential during the entire project.



Deliverable Lead	E-Seniors, End-User Organization
Document Status	
Internal Reviewer 1	Tim Dutz, TUDA
Internal Reviewer 2	Alexander Berman, TALK
Type	Deliverable
Work Package	WP2: Concept, Requirements and Specification
ID	D2.3: User Stories and Requirement Analysis
Due Date	28.02.2014
Delivery Date	27.01.2015
Status	For Approval

Document History	
Contributions	V0.1: ESE, CHA, NFE - 17.12.2013 V0.2: ESE, TALK, ASC - 02.01.2014 V0.3: CHA - 07.01.2014 V0.4: ASC, CHA, ESE - 08.01.2014 V0.5: CHA, ESE - 16.01.2014 V0.6: ESE - 23.01.2014 V0.7: ESE - 28.01.2014 V0.8: ESE, NFE - 06.02.2014 V0.9: ESE, IESE, NFE, CHA - 12.01.2014 V1.0: ASC - 13.02.2014 V1.1: ESE - 14.02.2014 V1.2: TUDA, ESE - 24.02.2014 V1.3: TALK, ESE, NFE – 26.02.2014 V1.4: ASC, ESE – 27.02.2014 V1.5: ESE – 27.01.2015: amended version after MTR
Final Version	January 27 th , 2015

Note

This deliverable is subject to final acceptance by the European Commission.

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Executive Summary

The main objective of this deliverable is to provide the ALFRED user stories that will be the basis for the ALFRED system architecture. These stories define both the functional and the non-functional requirements for the technical specification. In order to ensure that the ALFRED system meets the needs and demands of its intended users, the user stories are based on the findings from the ALFRED focus group sessions. Furthermore, an overview of existing user requirements in related projects enables us to verify the accuracy of our own findings as well as extend the information acquired from the ALFRED end-users.

The essential objective of the work in this deliverable is to investigate the users' opinions, needs and views. This process will lead to the verification of the ALFRED use cases and finally, to the definition of the ALFRED user stories under the four ALFRED pillars, namely:

- User-Driven Interaction Assistant that can interact with the user providing support to the daily activities.
- Personalized Social Inclusion element that prevents the users from the social isolation.
- Effective & Personalized Care element that allows monitoring the users' health status.
- Physical & Cognitive Impairments Prevention element that, by incorporating serious gaming, improves the physical and cognitive condition of the user.

The entire target group data found in this deliverable stems from the ALFRED focus groups, scientific literature, reports, studies and publications of related European funded research projects.

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1 Introduction

ALFRED – Personal Interactive Assistant for Independent Living and Active Ageing – is a project funded by the Seventh Framework Programme of the European Commission under Grant Agreement No. 611218. It will allow older people to live longer at their own homes with the possibility to act independently and to actively participate in society by providing the technological foundation for an ecosystem consisting of four pillars:

- **User-Driven Interaction Assistant** to allow older people to “talk” to ALFRED and to ask questions or give commands in order to solve day-to-day problems.
- **Personalized Social Inclusion** by suggesting social events to older people, considering their interests and social environment.
- A more **Effective & Personalized Care** by allowing medical staff or carers to access the vital signs of older people which are monitored by (wearable) sensors.
- **Physical & Cognitive Impairments Prevention** by incorporating serious gaming to improve the physical and cognitive condition by offering games and quests to older people.

1.1 ALFRED Project Overview

One of the major problems today is the increasing isolation of older people, who do not actively participate in society either because of missing social interactions or because of age-related impairments (physical or cognitive). ALFRED will allow overcoming this problem with an interactive virtual butler for older people, which is fully voice controlled.

The ALFRED project is wrapped around the following very clear main objectives:

- Empowering people with age related dependencies to live independently for longer by delivering a virtual butler with seamless support for tasks in and outside the home. The virtual butler ALFRED will have a very high end-user acceptance by using a fully voice controlled and non-technical environment.
- Prevailing age-related physical and cognitive impairments with the help of personalized, serious games.
- Fostering active participation in society for the ageing population by suggesting and managing events and social contacts.
- Improved care process through direct access to vital signs for carers and other medical stuff as well as alerting in case of emergencies. The data is collected by unobtrusive wearable sensors monitoring the vital signs of older people.

To achieve its goals, the project ALFRED conducts original research and applies technologies from the fields of Ubiquitous Computing, Big Data, Serious Gaming, the Semantic Web, Cyber Physical Systems, the Internet of Things, the Internet of Services, and Human-Computer Interaction. For more information, please refer to the project website at <http://www.alfred.eu>.

1.2 Deliverable Purpose, Scope and Context

The intention behind this deliverable is to ensure that the user requirements related to the ALFRED technology are analysed at an early stage and can thus be considered during the design and implementation phase. This deliverable will serve as a guideline for the

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ALFRED architecture definition (as described in deliverable D2.4). The use cases and the user stories defined in this report are essential for the project, as they serve as a guide for the project partners in order to develop a user-friendly and useful solution for the target groups.

1.3 Document Status and Target Audience

This document is listed in the Description of Work (DoW) as “public”, as it provides general information about the goals and scope of ALFRED and can therefore be used by external parties in order to get according insight into the project activities.

While the document primarily aims at the project’s contributing partners, this public deliverable can also be useful for the wider scientific and industrial community. This includes other publicly funded projects, which may be interested in collaboration activities.

1.4 Abbreviations and Glossary

A definition of common terms and roles related to the realization of ALFRED as well as a list of abbreviations is available from the supplementary document “Supplement: Abbreviations and Glossary”, which is provided in addition to this deliverable.

Further information can also be found at <http://www.alfred.eu>.

1.5 Document Structure

This deliverable is broken down into the following sections:

- Chapter 1 provides an introduction and outlines the purpose, scope, context, status, and target audience of this deliverable.
- Chapter 2 defines the methods that are used for the definition of the user stories, which is the main result of this deliverable. Firstly, this chapter describes the ALFRED pillars and the target groups, which are the basis for the organized focus group sessions. Secondly, the chapter gives an insight into the focus groups sessions organized in France, Germany, and The Netherlands, and then defines the methods used for the end-user data analysis.
- Chapter 3 gives an overview of the existing user requirements that have been defined in literature, studies, and projects that are related to designing ICT tools for healthy and active ageing. This overview enables us to put the user stories in a wider social context, which ensures that the project partners take into consideration the essential and pertinent user requirements.
- Chapter 4 provides the ALFRED use cases and personas that have been verified with the end-user data driven from the different focus groups. These use cases and personas will be the basis for defining the functional and the non-functional requirements.
- Chapter 5 defines the user stories (i.e., the functional and non-functional requirements) that will be an essential foundation to the definition of the ALFRED system architecture and that will allow for the specification of concrete tasks for the system development.
- Finally, chapter 6 concludes the findings of the deliverable.

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2 Methodology

The ALFRED project is applying a user centred design process to define the functionalities of the virtual butler. As part of this process, end-users are involved throughout the different phases of the project, starting with the definition of the user requirements, thus making sure that the results of the project align with the actual needs of older people and care systems in society. This chapter further explains the methods and procedures that were used to define the requirements, starting with the overall methodology, the focus group sessions and the baseline for the definition of the user requirements.

2.1 User Requirements Methodology

This section establishes the methodology for the collection of the user requirements in the ALFRED project. User requirements collection and analysis aim at providing insights about user needs related to the objectives of the ALFRED project.

The user requirements methodology that is used in the ALFRED project initially focuses on the definition of the target group, identifying who the actual end-users of the ALFRED system would be, and what their main characteristics are. These characteristics are further analysed with the help of existing research and literature, sketching a complete image of the end users in each of the pillars.

The target group is then visualized through personas. These personas reflect in a lifelike manner the characteristics of the target group of the ALFRED project. The use cases reflect the usage of the system by the personas. The personas and use cases that were defined in D2.1 'Strategy Consensus Document' are used as a basis during this process.

These personas and use cases are then verified, extended and enhanced through a set of focus group sessions, thus directly involving members of the target group in the project. The advantage of a focus group session is that older persons can exchange their ideas freely, which enables ALFRED's project partners to acquire a deeper understanding about its intended user groups, their needs and their ideas in regard to the ALFRED solution.

With the help of the focus groups, the personas and use cases are verified, enhancing and extending them to create a complete overview that describes the real-life sequences of events for a user using the ALFRED solution.

Finally, the use cases provide the basis for the definition of the user stories and the (functional and non-functional) requirements that form an essential input for the system architecture.

Figure 1 describes this process schematically.

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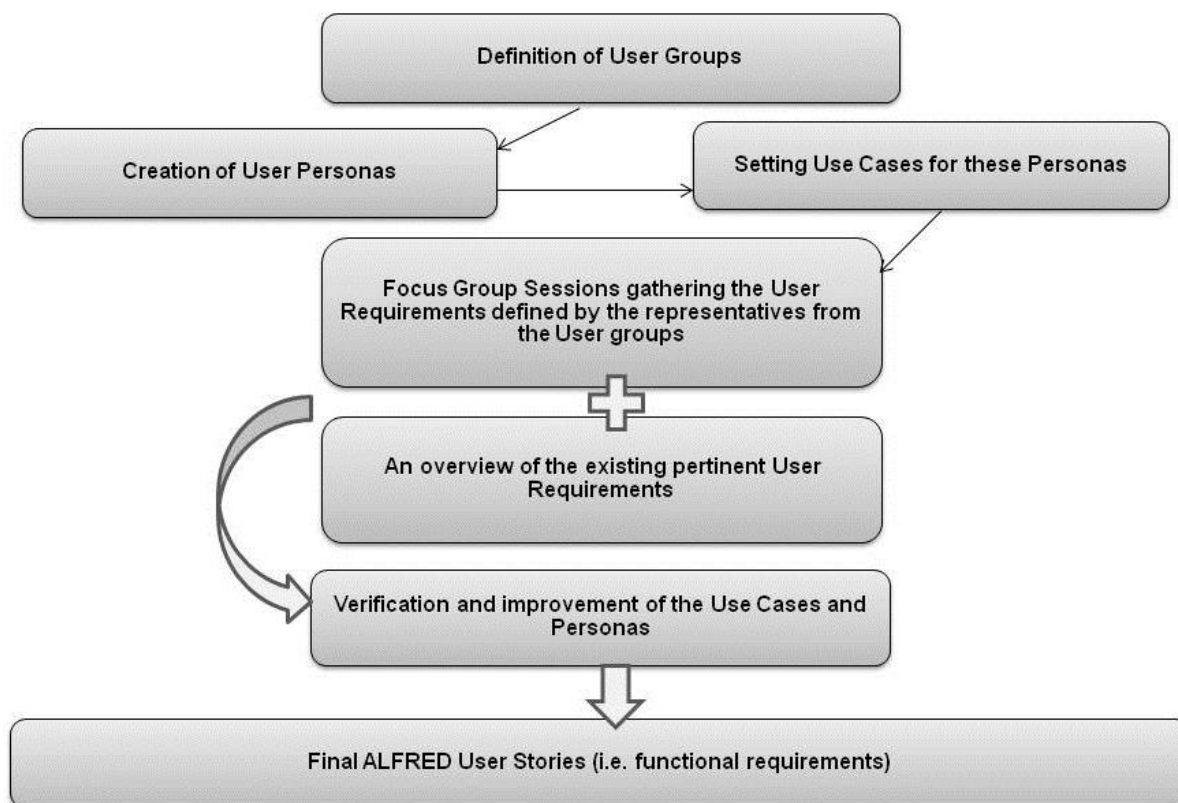


Figure 1: Illustration of the Different Work Steps within the T2.3

2.2 ALFRED User Groups

This section defines the user groups of ALFRED, that is, the people who will actually be using the results of the project. First, the three user profiles of ALFRED are described, where “the older adults” are the primary target group. The three user groups were enhanced through the focus group sessions, that and made more tangible with the personas in section 4.1. In this chapter, you will first have a general description of the primary and secondary user groups of the ALFRED solution, then (in the section 2.2.4) the more detailed segmentation of the different relevant target groups is given.

2.2.1 Older Adults

The primary end-user group of ALFRED consists of people aged 60 and above. This age threshold was chosen because the over 60-years-old persons are facing many changes in their lives that are related to 1) various physical and mental effects of ageing and 2) to the approaching retirement age. However, it has to be considered that ageing is a very complex process that affects different people in different ways. Persons from this target group very often display vast individual differences in physical and mental fitness as well as personal autonomy and therefore cannot be seen as a homogenous group. For individuals, old age is a relative concept. Many older people feel that they are treated in a patronizing manner, and therefore it is important not to use stereotypes and to not segment people based solely on their age.

Consequently, **attitude** and not age is the most fruitful approach to segmentation and targeting older adults. For this reason, ALFRED uses the “life stage model” as developed by the Center for Mature Consumer Studies (CMCS), which uses a person’s biophysical, psychological and social circumstances in life to describe an older person’s aging process [MOS96]. Hence ALFRED does not intend to put people in fixed boxes, but looks at stages of life and at the related needs in a specific stage to determine the requirements of the ALFRED target group.

People late in life can move from one stage to another. The changes between the stages may occur over time, or they may be abrupt. Furthermore, these changes are relatively age-indifferent, as they may begin at any time during one’s life, or never be experienced at all. Despite the expected physical and mental decay that usually comes with increasing age, both improvement and worsening of physical and mental conditions are possible in equal magnitude.

For example, one can consider a person that has been suffering from a long-term but surgically curable joint disease (e.g., arthrosis of the hip). This condition can certainly affect the physical ability to walk long distances and also cause the person a considerable amount of pain. The obvious effects are that the person would feel frail and isolated. The restricted sphere of action would make it harder to meet friends and family and can lead to social exclusion. The pain which is connected to this condition would make the person feel frail and the restricted mobility would lead to further decay of muscle tissue and a general physical degeneration of wide parts of the body system.

However, as with many age related conditions there are various medical treatments available that can prevent or even reverse this “downward-spiral” towards physical, mental and social decay. After a hip endoprosthesis, the person will relatively quickly be able to vastly increase her activity radius, reduce pain and build up physical strength, which would reverse the above mentioned effects of frailty, social isolation and overall decay. Many conditions that threaten elderly people to reduce their ability to function individually and live healthy and productive lives today can either be slowed down dramatically by preventive measures or reversed by various means of medical interventions. It should be mentioned that this is a complex multifactorial process that may require collaboration across professions. It also requires the compliance of the older person to address these age-related issues.

The bilateral model of Moschis [MOS96] is a good way to illustrate this process. In the first “healthy and social” stage in this model, people enjoy the benefits of being older, such as being retired, having free time, having financial stability and mental stability. People are healthy and independent and do not see themselves as older people in any way. The “frail isolated stage” can be seen as the complete opposite, where people have one or several psychical and/or chronic ailments and are also in isolation due to different circumstances in life. People in this stage are likely to think of themselves as being old.

Between those two poles, two more stages exist. When people encounter more psychological and socially related problems (a partner that falls away, a lack of social skills, depression) as they get older, they run the risk of becoming “socially excluded”. People are more socially withdrawn, although physically healthy and often secluded. When people encounter one or more physical problems, they enter the “physical impairments stage”. Although they have one or several chronic conditions, they have a high self-esteem and like to go out. They are worried about their physical independence

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and wellbeing, but adjust well to their frailties and try to make the best out of life. Figure 2 illustrates the different stages of Moschis' model. Specific arrows indicate that people may move to another stage due to physiological, psychological and social changes.

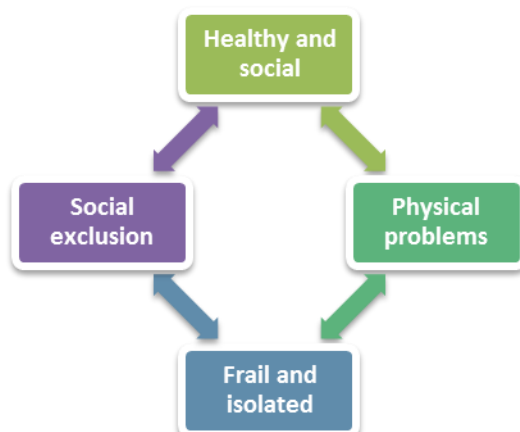


Figure 2: Life Stage Model by Moschis [MOS96]

Drawing on Moschis' model, ALFRED focuses on people that have needs related to social exclusion and physical problems, with the objective to prevent a shift to more isolation or frailty. ALFRED has the objective to prevent people from becoming frail and isolated by offering them support for social activities and physical and cognitive training and monitoring. ALFRED focuses on supporting people to become healthy and socially active and also has the goal to rehabilitate age related physical and cognitive impairments of older adults.

This definition of the target group here will not only focus the work of ALFRED, but will also be important in all dissemination and exploitation activities of the project towards older persons.

2.2.2 Caregivers and Medical Staff

Caregivers (both formal and informal) and medical staff form the secondary end user group of ALFRED. They will be interacting with ALFRED apps through specific interfaces and will directly benefit from the results of the project.

Informal caregivers are relatives, neighbours and friends who support a person in her activities of daily living. An informal caregiver does not receive any payment and is often a partner or child living in the same household. Informal caregivers will benefit from ALFRED solutions as they can stay up to date with real-time information about their loved one, which will hopefully grant them with a peace of mind. They will be able to use ALFRED to stay in touch and have direct contact with the older person in case of urgency. The remote interaction with the ALFRED user will allow carers a better coordination of care processes.

Formal caregivers can be social workers or carers, people that work for public or private organisations acting in the field of social wellbeing, home care, activity centres or charities. ALFRED will enable formal caregivers to take even better care of the older adults, reaching their clients better and offering more personalized support when needed.

Finally, doctors, nurses and other hospital staff will be able to provide more efficient care through direct communication with the patient.

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2.2.3 Developers and App Creators

ICT companies will benefit from the ALFRED project results as the project will provide an open platform that enables third parties to produce applications for the ALFRED system. During the course of the project, several of those apps for certain domains will be created, e.g. for scheduling appointments or recommending social events. In order to enhance the system further, external developers will be able to realise their ideas in order to extend and improve the users experience with the ALFRED system.

2.2.4 Specifications and ALFRED Target Group Segmentation

A special attention is given to the users' attitude toward ICT-solutions. In the wide target group of older people there are just as many attitudes towards ICT as there are ice cream flavours. In order to focus the project towards a specific target group, the attitude of an ALFRED user¹ towards ICT is described. The average ALFRED user (65+) has little or no ICT skills. The target user might be a bit hesitant towards using ICT solutions but can be reasonably easy motivated by their family, children or caregivers to use ICT. Many of the ALFRED users are currently using a senior phone: A mobile phone with very big buttons which allows them to make phone calls and send messages. They only use this senior phone when they are going on a trip somewhere. Due to the fact that they see their children or people on the street around them with smartphones, they are becoming more interested in additional possibilities a smartphone or tablet has to offer. A normal smartphone would be too difficult to use, as they are already experiencing quite often problems with their senior phone. They feel insecure about using a smartphone. They are unfamiliar with terms such as 'apps', 'widgets', 'wifi', '3G', etc. Therefore ALFRED is the perfect solution. ALFRED can introduce them to the countless possibilities ICT has to offer. As soon as they discover the added value for them they will be very motivated to use it to enrich and support their daily lives.

Table 1 describes the three different target groups of future ALFRED users. The "Primary target group" (PGT) is composed of older adults with varying degrees of independence or need for care. As in real life, negative effects on health conditions are likely to make people preoccupied with themselves and their own problems related to their state of health. For this reason we stated different attitudes to ICT in the PTG1, PTG2, PTG3 and the PTG4. The "Secondary target group" (STG) is composed of lay informal caregivers and formal caregivers. Informal care givers are laypersons such as family members or friends; they can be seen as a cross-section of the overall general population with various degrees of ICT skills, but at least basic abilities. Formal caregivers are members of any sort of health profession, due the need of using ICT in modern healthcare; they will have at least moderate ICT. App developers, able to create new service offer in the ALFREDO marketplace make the "Tertiary target group" (TTG).

Finally the table 1 refers to personas that are a design tool for technological projects enabling a user-centric approach for the system development (you can find more information about the personas in the chapter 4.1 and in the Appendix 1). As an example,

¹ When not otherwise specified the term "end user" is used for the primary target group of user, which is the older person who uses the ALFRED system with the app running on a mobile device to assist him during his day.

Olivia the Older person represents TG1” Independently living and autonomous older adults 60+” if she is in a good health, but she can also be considered in the TG2 “Independently living older adults 60+ having informal caregiver’s” after her sports accident when her spouse Carl the caregivers take care of her to get back in shape. With the same logic, Carl the Informal Caregiver can be represent the TG1” Independently living and autonomous older adults 60+” as he is an independent older adult, but when he is taking care of Olivia he is acting as a informal caregiver and as a result he can be also represent STG1: Informal caregivers.

Table 1: Target Groups of the ALFRED-solution

Target Groups	Target Group Category	Specific target group definition	Project tasks directly concerning this TG
Primary target groups (PTG)	PTG1: Independently living and autonomous older adults 60+	General: these persons pursue an independent lifestyle at home alone or with someone Attitude to ICT: some experience in certain ICT-tools or at least having an open attitude. Often they would like some support when using a new ICT-device for the first time. Once they have received the necessary support, the usage is autonomous. Chronic conditions: No chronic diseases but might suffer from age-related small physical problems, such as high blood pressure and cholesterol levels. Value added of ALFRED for this TG: helps these persons to stay independent and socially involved. Assisting them inside and outside home Persona representing this TG: Olivia the Older Person, Carl the Caregiver	T2.3, T8.1, T8.2, T8.3, T9.3, T9.6
	PTG2: Independently living older adults 60+ having informal caregiver's (friend, family member, spouse) support	General: these persons pursue an autonomous lifestyle at home alone or with someone but they have occasional support for their daily living, for instance to do their grocery shopping or preparing their meals Attitude to ICT: some experience in certain ICT-tools or at least having an open attitude. Often they would like some support when using a new ICT-device for the first time. Once they have received the necessary support, the usage is autonomous. Chronic conditions: No chronic diseases but might suffer from age-related small physical problems, such as high blood pressure and high cholesterol, reduced vision, reduced overall mobility, restrictions in range of motion, slightly reduced ability to perform activities of daily living and this reason they consult a doctor once in a while. Value added of ALFRED for this TG: helps these persons to stay independent and socially involved. Assisting them inside and outside home but also ALFRED enables them to be	T2.3, T8.1, T8.2, T8.3, T9.3, T9.6

		easily in touch with their informal caregivers and let them know how they are doing. Persona representing this TG: Olivia the Older Person	
	PTG3: Independently living older adults 60+ having formal caregiver's support	General: these persons pursue an autonomous lifestyle in a home environment but a formal caregiver give them regular support for their daily living, for instance monitoring their state of health or delivering meals. Attitude to ICT: some experience in certain ICT-tools or at least having an open attitude. Often they would like some support when using a new ICT-device for the first time. Once they have received the necessary support, the usage is autonomous. Chronic conditions: These persons might suffer from a chronic diseases such as diabetes, mild osteoarthritis, chronic Low Back Pain, diabetes and for this reason they benefit from a regular services of a formal caregiver Value added of ALFRED for this TG: encourage these persons to stay independent and socially involved. Assisting them inside and outside home but also ALFRED enables them to be easily in touch with their formal caregivers and let them know how they are doing. Persona representing this TG: Otto the Older Person	T2.3, T8.1, T8.2, T8.3, T9.3, T9.6
	PTG4 : Older adults 60+ with "Frailty-Syndrome"	General: these persons pursue an autonomous lifestyle in a home environment or in a hospital environment may have a formal or informal caregiver. because of chronic age-related reduced resilience. They also find it increasingly difficult to motivate themselves to participate in social events and to encounter interesting activities. Attitude to ICT: some negative effects of the chronic health condition may have an impact in their usage of the ICT-tools, but some experience in handling ICT-tools will most likely be present in most users appropriate familiarisation should enable them to use the system. Chronic conditions: These persons might suffer from chronic diseases such as coronary heart disease, diabetes, osteoarthritis, chronic obstructive pulmonary disease or current or past cancer and for this reason they benefit from having regular consultations in a hospital or with their doctor. Value added of ALFRED for this TG: ALFRED will allow these persons to be in contact with their entourage but also to monitor their physical wellbeing. ALFRED enables them to do physical exercise and to measure some essential vital data. ALFRED also can facilitate	T2.3, T8.2, T9.3, T9.6

		the contact with their entourage and health care providers Persona representing this TG: Hilde the Older Person	
Secondary target groups (STG)	STG1: Informal caregivers (such as friends, family members of the older adults)	General: these persons can be all ages and they take care of a close older adult who needs some support in the daily living and they do it as a voluntary activity. Attitude to ICT: These persons do not have a distinct age category as they are active and independent adults. They have at least basic skills with ICT-solutions, such as text editing on computer. Value added of ALFRED for this TG: ALFRED enables them to be easily in contact with the older adult that they take care of and to check that this person is doing well while they are not with them. ALFRED also helps in the time planning; they can use a shared calendar with the older person and set them important reminders. Persona representing this TG: Carl the caregiver	T2.3, T8.1, T8.2, T8.3, T9.3, T9.6
	STG 2: Formal caregivers (doctors, homecare nurses and other health care professionals of the older adults)	General: these persons are healthcare professionals and they work in different healthcare institutions. They take care of the physical and mental condition of the older persons. Attitude to ICT: these employees often use different ICT-tools and software in their work and have at least moderate ICT experience. These tools enable them notably to communicate and to check important data. Value added of ALFRED for this TG: ALFRED could help these persons to be more connected to the persons they are taking care of and enable them to check that the older adult is doing well. Furthermore, ALFRED will help them to offer more personalized support. Persona representing this TG: Mark from the Medical Staff	T2.3, T8.2, T9.3, T9.6
Tertiary target group (TGT)	TGT1: App developers	General: these persons are technology developers and they work in different private companies specialized in software development. These persons are looking for new markets and clients. Attitude to ICT: the developers are experts who develop new ICT-tools for clients. Value added of ALFRED for this TG: ALFRED will propose an open platform that enables third parties to produce applications for the ALFRED system. ALFRED helps them to reach new markets and clients. Persona representing this TG: Dave the developer.	T9.1, T9.3, T9.4, T9.5

In addition to the primary, secondary and tertiary target groups, the consortium has identified other relevant stakeholders that are described in the Table 2. These stakeholders, such as other research projects, industry actors or public structures, are identified in a this separate table as these not a part of the of “persona” methodology. These stakeholders will also be analysed in the other forthcoming project deliverables notably related to the Market and Applicability Watch Report (D2.2).

Table 2: Other relevant stakeholders of the ALFRED-solution

Other Stakeholders	Target Group Category	Specific target group definition	Related tasks
	Researchers and academic communities in the field of new technologies for active living and/or serious games	General: these academic stakeholders are active in the field of the assistive technologies for older persons and they are looking for new and innovative approaches. Attitude to ICT: these persons will have moderate of good knowledge about the ICT Interest for ALFRED: opportunity to promote actively and European widely the ALFRED innovations. Value added of ALFRED for this TG: ALFRED proposes an innovative and holistic approach of a mobile assisted living solution adapted for the older adults.	T9.3, T9.4, T9.7
	Other ongoing research projects in the field of new technologies for healthy and active ageing	General: These research projects are national or international and developing innovative technological solution for active ageing. Interest for ALFRED: an occasion to share know-how with relevant European stakeholders related to the ALFRED technologies and methodologies Value added of ALFRED for this TG: ALFRED will propose these projects a new scientific point of view and propose common dissemination activities in order to acquire a larger audience.	T9.3, T9.4, T9.7
	Industry: different sectors of industry are interested and are already involved or will be involved in the implementation of mHealth. They will naturally be interested in the outcomes of the project. Also, they can be ALFRED customers.	ICT and Telcos companies: ALFRED users will need a mobile device and an internet connection. While many might already have access to this, others do not. Collaboration and synergies of ALFRED with these companies shall be fostered.	T2.2, T9.1, T9.4, T9.7
		Pharma companies: ALFRED should develop synergies that can be fostered with Pharmaceutical companies, in particular in the	T2.2, T9.1, T9.4, T9.7

		management of chronic diseases (controlling diet, increasing physical activity, setting up reminders for medication)	
		Insurance Companies: health insurance companies are an important player in the mHealth market, and might be customers of ALFRED	T2.2, T9.1, T9.4, T9.7
		Care homes: in some situations, older persons that maintain their independence to some extent are living in care homes. Care homes are likely first entry markets for ALFRED.	T2.2, T9.1
	<u>Government bodies:</u> With health and wellbeing in the political agenda of all EU Countries, government bodies are very much involved as healthcare providers, policy makers and often as payers too. The involvement and collaboration with these bodies will be key towards the end of the project and in the implementation phase. Among EU, countries vary largely in the way their healthcare processes are organized, some are at Regional level, other at National. On top of that, there are EU Directives that apply to all countries.	Regional	T2.2, T9.1, T9.4
		National	T2.2, T9.1, T9.4
		European	T2.2, T9.1, T9.4
	<u>Community in General</u>	European citizens are interested parties in all EU Projects. Not only the funding comes from taxpayers' money, but ultimately citizens are the ones that will benefit from the project results. Therefore, it is fundamental to communicate and explain to a general audience the objective and importance of the project and the impact of its results	T9.4

2.3 ALFRED Pillars – Basis for End-User Discussions

The ALFRED system is formed by four core concepts, the pillars. During the user requirement gathering and analysis phase, each of these pillars is presented to the ALFRED target groups who will provide relevant feedback for each of the core concepts. These user stories will be used as a basis for the definition of the architecture of the ALFRED system.

2.3.1 Pillar I: User Driven Interaction Assistant

In order to empower people with age related dependencies to live independently for longer the ALFRED project will deliver a virtual butler with seamless support for tasks in and outside the home. This User-Driven Interaction Assistant is the first pillar of the ALFRED project and will be implemented as the ALFRED App, which allows users to ask questions and contact physical people by using their voice. In a sense, this functionality will provide a “Virtual Butler” to older people, helping them to get information, perform tasks and get in contact with other people.

2.3.2 Pillar II: Personalized Social Inclusion

In order to foster active participation in society for the ageing population the ALFRED platform will provide tools to suggest and manage events and social contacts. The services to provide such functionalities to the ALFRED platform constitute the second pillar of the ALFRED project.

2.3.3 Pillar III: Effective and Personalized Care

The aim of this pillar is to provide services to monitor the health status of a user. In order to achieve improved care process, vital signs will be accessible for carers and other medical staff. In addition, alerts can be sent in case of emergencies. The data is collected by unobtrusive wearable sensors monitoring the vital signs of older people. The services to provide such functionalities to the ALFRED platform from the third pillar of the ALFRED project.

2.3.4 Pillar IV: Physical and Cognitive Impairments Prevention

This fourth pillar of ALFRED focuses on mechanisms that motivate regular physical and/or cognitive activities in order to help reduce the effects of aging on the users, and it does this by way of so-called “serious games”. Serious games are games that strive to have a positive effect on their players besides mere entertainment. They use game mechanics to motivate the user to perform a task that she would otherwise consider dull or tedious, such as learning new skills or working out.

2.4 User Requirements Assessment Methodology

The end-user partners involved in task 2.3 organized several focus group sessions with representatives from the ALFRED user groups in order to define their needs and requirements related to the ALFRED technology. The collected end-user data was analysed, the ALFRED use cases were verified and enhanced and user stories were defined on the basis of this data analysis. This chapter details how the focus group sessions were organized and how the gathered end-user data was processed.

2.4.1 Focus Groups sessions

A total of six separate focus group sessions were carried out in conjunction with task 2.3. Each of the three end-user organizations (partners ESE, OUD, and CHA) carried out two focus group sessions related to “their” respective pillar of ALFRED. Each focus group had four to five participants (older persons above the age of 60 with different backgrounds) as well as one observer from the end user organization. The focus groups discussion were

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organized by the end-user organizations of the project: by E-Seniors (ESE) in France, by the National Foundation for Elderly (OUD) in The Netherlands, and by Charité Department of Geriatrics (CHA) in Germany. The end-user organizations divided the ALFRED pillars according to their expertise in order to get a better focus on the end user involvement.

In addition to the older adults, professionals from different domains related to the relevant pillars were invited to take part in the focus group sessions (such as IT experts, logopedics, caregivers, business developers, and wellbeing organizations such as Red Cross). These are people who have an expert view and can see the wider scope of ALFRED.

2.4.2 ALFRED Focus Group Participants

In total, 30 participants that where older adults and formal or informal caregivers took part in the ALFRED focus groups. All of these participants were French, Dutch, German or Italian. They had very varied levels of education (going from Secondary school until PhD degree). Table 3 describes the focus groups in terms of number of participants, age and gender distribution and their usage habits of different ICT-tools.

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Table 3: Summary of the ALFRED Focus Group Participants

	France (ESE)	Netherlands (NFE)	Germany (CHA)
Number of organized Focus Groups	2	2	2
ALFRED Pillars discussed in the sessions	Pillar I Pillar IV	Pillar I Pillar II	Pillar III Pillar IV
Total number of participants	10	10	10
Number of older adults involved	9	9	6
Number of formal and informal caregivers involved	1	1	4
Number of Female and Male participants	9 (F) and 1 (M)	7 (F) and 3 (M)	5(F) and 5 (M)
Average age of the participants	71 years	76,5 years	62,5 years
How often the participants use of ICT-tools (i.e. computer, laptop, smartphone or tablet)?	10 persons use daily	10 persons use daily	8 persons use daily, 1 person uses weekly, 1 person rarely
For which activities the participants use the ICT-tools?	Mail, news, communication, work, games, videos, geolocalisation, bank services	Mail, news, communication, work, bank services, information research	Mail, news, communication, work, games, music

2.4.3 Analysing the Data Derived from Focus Groups

The following process, which was inspired by an academic study [HD06], has been used for the qualitative content analysis:

The first step consists of audio taping and making field notes during the interviews with the main goal of recording the impressions of the interactions between the participants rather than focusing on plain word-by-word note-taking of the spoken content.

This is followed by step two, in which reflective journalizing immediately after the group discussion is performed. As long as the memory of the moderator is still fresh, the main ideas, concepts or issues raised by the participants should be documented.

The third step consists of listening to the audiotape and if necessary to do amendments to the previously written field notes. This step is necessary to ensure that the field notes are a precise reflection of the interaction and may require listening to the audiotape several

times. The transcriptions from all the ALFRED focus groups can be found in the annexes (Appendix 2) of this deliverable.

The fourth step is a preliminary content analysis in order to define first common themes. The predefined “pillar-specific” core issues were used for the focus group data analysis, enabling an easy orientation in the definition of the user stories. These themes will be further refined and categorized in order to verify and improve the ALFRED use cases and personas as well as to produce the “user stories”.

As a last step, a thematic review of all the analysed interview data is performed in order to create the final user stories which are then grouped into different themes.

3 Overview of the Existing Requirements

In the field of active ageing many ICT projects have performed research on user requirements. This section will give a short overview of the main conclusions of these different user requirement analyses, giving important insights for the ALFRED development process.

3.1 Active Ageing and ICT

The use of ICT can assist active aging and, particularly, help social inclusion and efficient care as well as stimulate physical and cognitive capacities of the user. Furthermore, in today's digital society, more and more older persons take the advantage of using ICT tools in order to facilitate their everyday lives.

Besides radio and TV consumption, accessing the Internet is the most visible aspect of ICT within the community of the older adults. According to a study published in 2007 by the University of Jyväskylä in Finland [JKN07], older adults, aged between 55 and 90 years, use the Internet mainly for communication, online services and information search, similarly to the other age groups. The results also suggest that age, education and place of living are significant factors in determining whether an older person makes use of ICT. Figure 3 illustrates the Internet usage habits of older persons in the European countries.

However, usability and accessibility issues as well as age-related physical restrictions can set severe barriers to technology use among elderly persons, and these barriers can even lead to digital exclusion. The involvement of the target users in the development phase of the technology is a suitable approach in order to consider ways for overcoming these barriers, which are of three main types: physical, cognitive, and economic.

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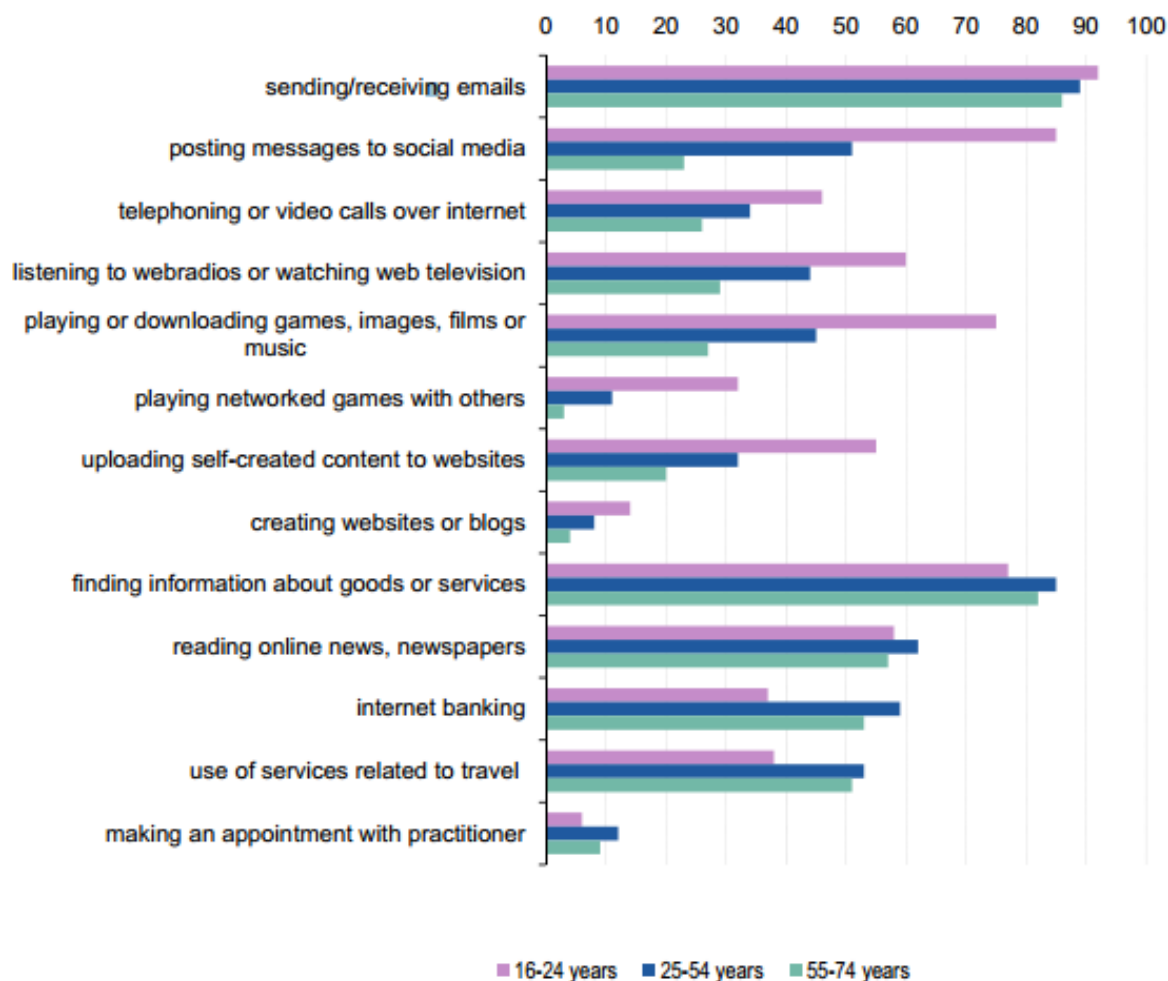


Figure 3: Internet Usage Habits (% of Internet users) (Eurostat, 2012²)

3.1.1 Physical Barriers

Age-related physical impairments can be a real barrier for older persons when it comes to usage of digital devices such as smartphones or tablets. Visually related difficulties among aging adults include problems with scrolling texts (due to the scrolling speed), with seeing in dim light, with reading small fonts, and with locating objects. Hearing impairments among older adults are widespread but often moderate or mild, and are expressed as noise sensibility as well as impairments with certain frequencies, speech recognition and sound localisation [APA98]. Finally, older persons may have mobility and coordination impairments that can make them reluctant to do certain movements: a lack of balance, coordination and reflexes [NIH07] can discourage them to perform certain activities or use mobile products that often require accuracy and speed of dexterous movements. Despite a distinctive interest in being active, elderly are also concerned about finding the right balance between activity and rest throughout the day [A2E2].

² Eurostat: "Internet use in households and by individuals in 2012":

http://epp.eurostat.ec.europa.eu/cache/ITY_OFFPUB/KS-SF-12-050/EN/KS-SF-12-050-EN.PDF

3.1.2 Cognitive Barriers

Physical limitations are interrelated with mental barriers that discourage older persons from trying out new things or learning new skills, which also influences their uptake of new technologies. It may take more time for older adults to encode, store, and retrieve information. The rate at which new information is learned can be slower among aging adults, which may lead to low self-esteem. Long-term memory shows substantial changes with age, while short-term memory shows less age-related decline. Attention and split attention capability is reduced and automated responses may be slow to learn and more difficult to suppress. On the other hand, users maintain well-learned skills and abilities, such as reasoning, well into old age [HAI93]. Furthermore, wisdom and creativity often maintain until the very end of life [APA98].

To this end, the developed technologies should provide solutions reducing the mental barriers related to the use of ICT tools. Notably, simplicity in the usage is an essential character of a solution that is targeted for older persons. Furthermore, technical support should be guaranteed in case of questions or problems with using the corresponding technologies, which will reinforce the reliability of the device.

3.1.3 Other Factors Influencing Technology Uptake

In addition to physical and cognitive barriers, economic constraints may also discourage older adults to use ICT devices. This situation is likely to vary between countries in Europe and generally, the cost of ICT tools and Internet access has fallen in recent years [BS08]. As a consequence, special attention should be paid to pricing policies and payment schemes of the developed solutions.

There is also a set of other factors influencing the technology uptake, that relate mainly to the personal situation of the older person. Data privacy issues should be considered at early phases of the solution development. Two key aspects are especially important: security, guaranteeing the protection of personal data, and control, assuring that the user can influence how her personal data may be used.

Perhaps most importantly, one should always bear in mind that the older end-users do not want to be considered as frail and at a disadvantage.

Table 4 summarizes the main factors influencing technology uptake by older people as pointed out in this section.

Table 4 – Factors Promoting Technology Use [NSG13]

Factors influencing technology uptake	Factors influencing the need for technology	Factors favouring technology uptake
Cost	User generation / cohort	Provide a clear additional value
Compliance with individual needs	Housekeeping style	Easier to use than expected
Previous personal experience with technology	Number and type (partner, children) of inhabitants in house	Pleasing user experience

Accessibility barriers (cognitive and physical)	Personal attitude towards technology	Adequate privacy of data
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3.2 General End-User Requirements

Given some of the general and more specific impairments that are related to ageing, adaptability of the technology to the end-user capacities, both physical and cognitive, is fundamental to ensure adherence, comfort and beneficial use of the ALFRED solution.

The user interface will be the point of interaction of the older users to assist them in the monitoring of their daily health and in social activities. It should be engaging and user-friendly to ensure usability. The target population requires specific considerations regarding [RC04, HDKAE09]:

1. Visual elements
 - Design a layout that is simple and consistent
 - Ensure that colour and contrasts make it easy to distinguish and read icons and texts
 - Choose appropriate police and font sizes so that the text is readable
 - Consistent font sizes along the application
 - Select an adequate screen size
 - Adapt the speed of interaction
 - Allow for personalization of the visual interface
2. Auditory elements
 - Use an appropriate frequency to deliver messages
 - Allow for personalization of the voice
 - Speak the “language” of older people, plain language, no jargon or technical terms
 - Use the appropriate temporal and spectral characteristics for the speech recognition
3. General Usability Testing
 - Provide general simplicity in usage and learnability
 - Ensure that the “back” button behaves predictably
 - Make it obvious what is clickable and what is not
 - Have access to consistent and ongoing technical assistance and support
 - Be constituted of integrated accounts to avoid entering many passwords
 - Allow easy customization according to the preferences of the user
4. Confidentiality & privacy
 - Ensure confidentiality and protection of personal data (to counteract fear of losing control of personal data)
 - Let the user stay in control of personal data

3.2.1 Empowering People with Age Related Dependencies

Many older people want to live independently and to stay at their own homes. The objective of the ALFRED project is therefore to extend the independent living of older people and improve their quality of life while providing a level of safety that is equivalent to

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the one they would have within residential homes. The user driven technology assistant will allow older people to talk to ALFRED, the virtual butler, and may be of great use to assist those with age related dependencies: providing a reminder for medication, scheduling an appointment with their doctor, helping the person to go home, etc. The performance of these tasks should be simple and safe for the users inside and outside home. The ICT-tools for older adults should [RC04, GSI10]:

- Ensure protection from personal injury, falls, safety while performing tasks
- Precise localization (error margin should be minimal to avoid confusion)
- Avoid redundant and repetitive tasks when not required
- Use conventional interaction elements
- Alert a caregiver if abnormal behaviour is detected

3.2.2 Fostering Active Participation in Society for the Ageing Population

In addition to allowing users to ask questions to ALFRED, the system will also actively contact its users and “talk” to them from time to time, e.g. to suggest social events. This will mainly be done in order to improve the active participation of older people in society. . The ICT-tools for older adults should [MB10]:

- Facilitate live interaction
- Facilitate the organization of face-to-face contacts (to counteract the fear that they will have less physical contacts because of using remote communication devices)
- Maintain contacts with younger family members and re-establish links with lost contacts
- Facilitate participation to community events
- Propose activities that are financially suitable
- Propose activities that fit the routine of older users (times for meals, day care, etc.)
- Propose learning activities that match the personal interests of the users
- Provide adequate and timely notifications of online and offline events
- Explain why some personal or sensitive information are required for some service (to preserve the sense of privacy)

3.2.3 Improved Care Process through Direct Access to Vital Signs

The ALFRED device will support carers and medical staff to achieve more effective and seamless care. For achieving this, the ALFRED solution will know about the contact points of the user and will allow carers and medical staff to interact with the ALFRED installation of the user and access information about the elderly user as perceived by (wearable) sensors. Monitoring of vital signs should [GSI10]:

- Require minimal and very simple interaction of the user with the remote monitoring (e.g. turn on-off)
- Have minimal impact on the patient
- Not interfere with her daily routine
- Be available at home and on the go
- Be integrated in clothes or devices which are actually wearable (comfort and design)
- Provide access to a carer or to information about your health at any time

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3.2.4 Preventing Age-Related Physical and Cognitive Impairments

From time to time (or if explicitly requested by the user) ALFRED will offer to play games with the user. Those games are not intended to simply keep the person busy, but to help improve her physical and/or cognitive condition. Such games should [GA08, Mol14]:

- Allow others to join the game remotely in order to encourage social interactions
- Facilitate intergenerational interactions, such that for example grandfathers/mothers can play games with their grandchildren
- Encourage mild physical activities, like going for a walk
- Also work for people who do not like playing games in general
- Implement reward schemes to encourage regular gaming and improvement of performance
- Be personalizable to a certain degree to better adapt to the specific user

4 ALFED Personas and Use Cases

4.1 ALFRED Personas

Personas are a design tool for technological projects [COO99]. The aim of the persona method is to raise empathy for the end users in a development team as well as a deeper understanding of their drives and motivations. Personas allow developers to define the users they for which are developing and designing products. The personas are fictitious persons with their own characteristics including age, gender, state of health, attitudes etc.

In ALFRED, the personas that were defined in deliverable D2.1, have been further developed with the help of the CURE Elderly Personas database framework. The CURE Elderly Personas are synthetically generated from average traits mixed across countries³. They do not represent any private data from a single person and do not infringe any privacy and data security rights but use data from SHARE (project QLK6-CT-2001-00360). These personas have been enriched with information from the focus groups. Based on the focus groups results, an additional persona was created, reflecting a user with a larger support need.

The personas consist of six different user profiles that have been defined as stereotypical users of the ALFRED technology. These detailed description tables can be found in appendix 1.

4.2 Use Cases

Based on the focus group results and the adapted personas, the use cases that were described in the deliverable D2.1 “Concept, Role Definition and Strategy Consensus” have been revised, enhanced and enriched. The use cases now reflect more lifelike scenarios of use of ALFRED. The use cases are the basis for the user stories that will be further detailed in section 5. Each use case reflects a series of user stories that are mentioned below.

Table 5: Use Case – 1.1: Set Up with Care Organization

Use Case ID:	1.1
Use Case name:	Set up with Care Organization
Personas:	Otto the Older Person and Mike from the Medical Staff
The Home Care organization is implementing ALFRED as part of its new innovation programme, offering better and more efficient services to its clients. As part of the programme, Otto the Older Person receives a smartphone from the organization, which comes with ALFRED pre-installed. The costs of ALFRED and the smartphone are covered by his health insurance package. Mike from the Medical Staff brings Otto the smartphone and together they set up the user profile of Otto. Otto is somewhat insecure about security and wants to know exactly what ALFRED does with his personal data and if it shares information about himself only with his specific permission. After they have set up a plan on who can see which part of Otto's data, Otto is happy to indicate his hobbies and interests and he lists his contacts, namely his friends and neighbours. Otto also wants	

³ CURE-Elderly-Personas, Manual [Online] - 03 08 2012 - <http://elderlypersonas.cure.at/>.

to strengthen his social network and be able to share information on social events with other users, as he is feeling increasingly isolated. Together they set up a profile on ALFRED that allows Otto to connect with other ALFRED users so Otto can be notified about social, sporting and cultural events in his surroundings.

During the setup process, there are also some questions regarding his health status as well as his medication. Otto is physically healthy and does not need much medical attention. But he is worried about needing more support as he gets older. Therefore Otto agrees to set up a basic health profile together with Mike on ALFRED to share some information on his heart rate and emotional status that later can be expanded. Apart from the setup, the only technical task that Otto has to take care of is to frequently charge the battery of the smartphone that ALFRED is running on. All other interaction with the system is done simply by voice.

Table 6: Use Case – 1.2: Personalizing ALFRED

Use Case ID:	1.2
Use Case name:	Personalizing ALFRED
Personas:	Otto the Older Person
The first few days Otto the Older Person is getting acquainted with all the services ALFRED has to offer. For Otto it is easy to start to use ALFRED as the solution introduces himself to Otto and show the different functionalities that might be helpful for him. Otto is confident knowing that the device does not do any activities without his preliminary permission, as he has verified that this setting is on. This is important to him as he wants to keep the control over the device. Through the initial training, Otto gets acquainted with the agenda options, where he can add new appointments by simply speaking to the device. ALFRED reminds Otto of his agenda entries on a daily basis. ALFRED suggests that he could share his agenda with Mike from the Medical Staff, but Otto refused this as he considers this kind of information private. ALFRED is up to date on Mike's visits to Otto, so he gives him a short notice when Mike is coming so that Otto does not get impatient while waiting. Whenever ALFRED reminds Otto of an appointment, it suggests navigating him there. By inserting the information in the agenda, ALFRED also learns more about Otto and can relate his contact persons with the items in his agenda, as well as the locations to navigate him there. ALFRED reminds Otto every day to take his medicine. Sometimes Otto is confused and asks for a clarification. ALFRED then explains which medicine it refers to and what the box looks like. In case of any important health-related questions, Otto can easily consult Mike with ALFRED.	

Table 7: Use Case – 1.3: Personalized Social Inclusion

Use Case ID:	1.3
Use Case name:	Personalized Social Inclusion
Personas:	Otto the Older Person
ALFRED suggests different activities to Otto the Older Person on a daily basis. Often Otto does not feel like it, but sometimes his interest is caught. Last week, ALFRED mentioned that at the local activity centre they are organizing a pool championship. This is something Otto really likes so he decides to have a look. Otto enjoyed the event and met some people that were interested in setting up a small pool club,	

so he added their contact details and they arranged to meet the next week at the activity centre.

Sometimes Otto prefers to chat with his friends from home, and he has discovered that with ALFRED, he can have video calls that amuse him as he can see the person with whom he is speaking.

Table 8: Use Case – 1.4: Effective and Personalized Care

Use Case ID:	1.4
Use Case name:	Effective and Personalized Care
Personas:	Otto the Older Person and Mike from the Medical Staff
For a couple of months now, Mike from the Medical Staff has been taking care of Otto the Older Person. Otto is an uncomplicated patient, who does not have any severe problems besides his blood pressure, which is a bit high. Mike regularly visits Otto to monitor his blood pressure. Twice a week, Mike calls Otto on the phone to ask him about his health status. However, there are times when Mike is not able to visit Otto as often as he needs to and consequently, he is sometimes concerned about Otto's wellbeing. Therefore he talked to Otto about his concerns and they both agreed it would be a good idea to have the ALFRED device to be included in the care process in order to improve Otto's healthcare and safety. The ALFRED solution seems a good option for enhancing the care process as its functions are expandable to the user's needs. Furthermore, for this purpose Mike hands a new shirt to Otto, which contains a set of unobtrusive sensors. These sensors can send Otto's vital parameters via ALFRED to Mike. This way, Mike is able to keep track of Otto's health. Mike and Otto agreed that this solution would benefit both of them. However, Otto is also concerned about his privacy and feels that it is very important that he can decide which of the sensor data is available to Mike. He also wants to be able to check the vital data himself. Otto also wants the new shirt to look nice and fashionable and does not want it to make him look like a patient. Otto and Mike have also agreed that not all of the functions of the shirt have to be enabled at all times. However, Mike has informed Otto the health data that these sensors record can provide information about why he is feeling pain or stress. Mike and Otto agreed that it would be good if ALFRED could remind Otto to take his pills on a regular basis and Otto confirms to ALFRED once he has taken his pills. Furthermore, ALFRED should be able to collect and/or keep track of Otto's heart rate, measure his body temperature, heart rate, breathing frequency, activity rate and the amount of taken steps. Mike is aware that Otto does not meet a lot of people, and so he wants to make sure that Otto is in a good shape and that he treats his health problems. Sometimes Mike call to Otto with the video call in order to check that he seems well.	

Table 9: Use Case – 1.5: Games for Cognitive Stimulation and for Staying Fit

Use Case ID:	1.5				
Use Case name:	Games for Cognitive Stimulation and for Staying Fit				
Personas:	Otto the Older Person and Mike from the Medical Staff				
When Otto the Older Person feels bored, he asks ALFRED what to do and it might suggest a fun game. ALFRED offers a good selection of games that stimulate both the body and the mind. Otto knows that it is important to play these games on a regular basis, since Mike from the Medical Staff recommended doing so. Otto can also play together with a friend over the Internet, which Otto					
D2.3_User_Stories_Report_V1.5 (after MTR).docx		Document Version: 1.5	Date: 2015-02-03	Status: For Approval	Page: 35 / 115
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appreciates as it makes him feel less lonely.

The mind stimulating games enable Otto to keep his brain active. Otto can sometimes be a bit impatient and this is why he does not like to play one game for too long. Luckily, ALFRED also proposes mind stimulating games that are short so that Otto does not have the time to become frustrated while doing his memory exercises.

When Otto feels that he has not moved enough during the day, he plays games that make him perform mild physical activities. These games set specific goals for Otto, which motivate him to do all the proposed exercises. In the end, when he sees the results of the exercise (for instance, how many calories he has burned or which heart rate he had while playing), he is satisfied.

Mike is very satisfied with the ALFRED games that Otto plays weekly, as he can observe Otto's performance and easily check if his patient is feeling well. Mike is also able to coach Otto to perform better. Furthermore, Mike can directly ask Otto what is going wrong if he sees that Otto has not played for days or if he has very weak results. With Otto's agreement, Mike can ask ALFRED to send a regular summary of Otto's exercise sessions.

Table 10: Use Case – 2.1: Set Up by Older Person

Use Case ID:	2.1
Use Case name:	Set up by Older Person
Personas:	Carl the Informal Carer, Olivia the Older Person
Since Carl the Informal Carer was worried about his wife Olivia the Older Person, he has been looking for a solution that can both help Olivia to get better and also give Carl some more support in his care for her. He heard about ALFRED through a friend and he got interested as ALFRED is available on mobile devices, which makes it usable both indoors and outdoors. He bought two packages. Each package consists of a smartphone with the ALFRED app pre-installed, an Internet flat rate plan, and a simple step-by-step manual. Furthermore, Carl is satisfied as an after-sales service is provided in case he has trouble with the ALFRED device. Together, Carl and Olivia configure the ALFRED apps. Carl assigned the role of informal carer to himself. Additionally, they took a look at the easy-to-use ALFREDO marketplace for additional apps providing functionalities that are interesting for their specific situation. Based on their initial choices, ALFRED made more suggestions for other apps. Carl and Olivia are convinced that they are motivated to use ALFRED on a daily basis, as they can customize the ALFRED functionalities according to their preferences, which will enable them to use only the most useful and interesting services.	

Table 11: Use Case – 2.2: Physical Impairment Rehabilitation

Use Case ID:	2.2
Use Case name:	Physical Impairment Rehabilitation
Personas:	Olivia the Older Person and Carl the Informal Carer
Olivia the Older Person broke her leg few weeks back in a skiing accident and now she uses ALFRED to do her rehabilitation. ALFRED reminds her to start the exercises of the rehabilitation and tells her exactly what she has to do. When she asks for clarification, ALFRED shows her a video with an example. Olivia is motivated to use ALFRED for her rehabilitation as ALFRED sets different goals for each week, and after a few weeks of regular physical exercise with ALFRED,	

she can observe the results: her broken leg is getting back in shape.

Sometimes Olivia plays a game over the Internet together with Carl the Informal Carer while Carl is away traveling for his job. This enables them to spend some time together, and also allows Carl to check that Olivia is doing well while he is away.

Table 12: Use Case – 2.3: Informal Care Coordination

Use Case ID:	2.3
Use Case name:	Informal Care Coordination
Personas:	Olivia the Older Person and Carl the Informal Carer
Every morning ALFRED asks Olivia the Older Person how she is feeling. If ALFRED learns that Olivia does not feel well, it informs Carl about this. This is very useful for Carl who cannot be with his wife all the time even though she has health issues. ALFRED also indicates to Olivia when she has to take her medicine and Olivia confirms this to ALFRED as soon as she has taken it. Carl can check this by asking ALFRED if Olivia took her medicine. This way, Carl does not have to annoy Olivia by asking her, and he feels more at ease. Thanks to ALFRED, when Carl is not at home, communication between Carl and Olivia is now easier: they can send little messages with ALFRED. Because Carl is busy with work and taking care of Olivia, he uses ALFRED to remind him about things that he should do at home. During her days, Olivia enjoys cooking and with ALFRED she can even discover healthier recipes, as eating well is an essential part of wellbeing. In urgent situations, Olivia can ask ALFRED for help. ALFRED then checks on which of her carers are close by and calls them to check up on Olivia. This is also a great relief for Carl, who has been feeling much less stressed lately.	

Table 13: Use Case – 2.4: Physical Impairment Care

Use Case ID:	2.4
Use Case name:	Physical Impairment Care
Personas:	Olivia the Older Person, Mike from the Medical Staff

After her skiing accident, Olivia the Older Person is released from hospital and has to start her rehabilitation process for her broken leg at home. Twice every week Mike from the Medical Staff comes to visit Olivia to check on the condition of her foot.

Olivia discovered that she can use ALFRED in her care process and it can help her to get back in shape as ALFRED offers different kinds of serious games that train her muscles, movements, balance and endurance.

Since Olivia has to perform physical exercises, she will need her hands free and have a big screen in order to execute ALFRED's physical exercises in a safe manner. Mike has pre-selected a set of specific exercises from the ALFRED device in order to support Olivia's care process. In order for Olivia to stay motivated, the games have to have something in common with her favourite sports, such as a "skiing-theme" or "winter-theme".

Furthermore, in order to avoid boredom and keep motivation high, ALFRED suggests different games, for example one day balance games, another day games for strength and the next day games to improve mobility.

ALFRED will also remind Olivia to maintain a good body posture and will frequently give her tips on how to lead a healthy life. Since Olivia now spends a lot of time at home, ALFRED can provide her with some fun and creative games that she can play from her chair. She can also play together with a friend over the Internet, and together they can pass long moments playing. At any time, Olivia can also observe her game results and compare them to those of her friends.

Table 14: Use Case – 3.1: Adapted Interaction and Support for Daily Activities at Home

Use Case ID:	3.1
Use Case name:	Adapted Interaction and Support for Daily Activities at Home
Personas:	Hilde the Older Person
Hilde the Older Person received the completely installed ALFRED device from her daughter. She is not very used to new technology and she often needs help when using ALFRED. Luckily, ALFRED can provide her with instructions for usage and if she still requires assistance, ALFRED can automatically call her daughter. Hilde has customized the signal tones in ALFRED so that she can easily hear those. Particularly, the ALFRED vocal interaction voice speaks clearly and slowly enough, which enables Hilde to understand what ALFRED says. Furthermore, as Hilde has trouble seeing small written texts, she has adapted the font size of her ALFRED apps when she needs to read texts, which happens quite rarely. She likes that all the buttons in ALFRED are big enough to see and that she can manipulate them easily. She really likes the vocal interaction mode with ALFRED that she activates with a specific button, as this mode enables her to speak with ALFRED instead of reading and writing text that would take much more time. With the vocal interaction mode, Hilde can use the different ALFRED functionalities with her speech. For instance, ALFRED can ask her questions, such as would she like to go out and do something, if yes, ALFRED could propose outdoor activities in the neighbourhoods. These spontaneous questions with ALFRED amuse Hilde and give her new activity ideas. She turns off this ALFRED question mode, when she wishes to stay tranquil.	

Table 15: Use Case – 3.2: Support Activities on the Road

Use Case ID:	3.2
Use Case name:	Support Activities on the Road
Personas:	Hilde the Older Person
When Hilde the Older Person goes outside she carries ALFRED with her. Because it is so small and easy to take with her, she never leaves it at home. ALFRED keeps her up to date on her voluntary work, which motivates her to participate more. She is also less afraid to go outside as ALFRED can detect any dangerous situation, such as a fall and notify her daughter. Also when she gets lost she can easily navigate by using ALFRED, hopping on the bus if she gets too tired to walk. This really increased her confidence to go out and move about. Hilde has given ALFRED permission to propose different activities in her town, and she is particularly interested in exhibitions and museums. At the same time, ALFRED gives her daughter more peace of mind as she receives a notification when Hilde strays away too far from home. She can then easily call Hilde, or she can locate her by checking her location with the help of ALFRED. Hilde is also reassured as she knows that ALFRED can call her a taxi if ever she gets lost and too tired to walk back home.	

Table 16: Use Case – 3.3: Cognitive Impairment Rehabilitation

Use Case ID:	3.3
Use Case name:	Cognitive Impairment Rehabilitation
Personas:	Hilde the Older Person, Mike from the Medical staff
Hilde the Older Person now spends a lot of time at home and due to her health and mobility restrictions, she has a lot less exposure to the outside world than she used to have some years ago. Apart from her physical limitations, Hilde's daughter also realized that in the last couple of months, Hilde has been becoming more and more forgetful. She sometimes cannot remember names of friends anymore and has trouble in dealing with numbers. This conditions increasingly worried Hilde's daughter. Together with Mike from the Medical Staff who already checks on Hilde's cholesterol, they went to a medical centre to check on Hilde's cognitive abilities. The tests confirmed that Hilde suffers from a mild cognitive impairment. A condition that does not require pharmacological treatment, but that can be potentially improved by having increased cognitive input. Mike uses the ALFRED device already with other persons who have trouble remembering things. Together, Hilde and Mike agreed to use ALFRED for playing some easy serious games to increase her cognitive workload and improve her mental capacity. For Hilde it is important that she can use her imagination and be creative in the ALFRED games. She also would like to see some of her personal interests and hobbies such as music and arts to be represented in the games. Hilde liked to be outdoors a lot when she was still healthy but now that she is more restricted to the house, she is keen to play games that take her outside. Mike also suggested that some small physical games between the cognitive sections will be good to improve her memory, especially movement that involves a cognitive component such as opposite arm and leg coordination. Both Mike and Hilde also like that the ALFRED games have scores that show if Hilde is improving when she plays the games. In the future, Hilde also wants to play together with a friend over the Internet, and together they can pass long moments playing.	

Table 17: Use Case – 4.1: ALFREDO Marketplace

Use Case ID:	4.1
Use Case name:	ALFREDO marketplace
Personas:	Dave the Developer
Dave the Developer has convinced his manager to develop extensions for ALFRED, so he started developing an app that can read data from the health sensors a user is wearing. Dave starts his work by studying the API documentation and getting familiar with features ALFRED provides (e.g. components regarding speech recognition). He also has a look at the ALFREDO marketplace in order to check the terms and conditions regarding the sale of his applications. Dave steps through the API documentation and code samples in order to find out how he can define questions and actions the user can use to interact with ALFRED. He finds out that defining such questions is pretty simple and starts defining such parameters for his purpose. The ALFRED platform documentation also contains some rules an app developer has to follow in order to suit the ALFRED concept (concerning design and permissions the app will be granted). As soon as Dave has finished the development and testing of his app, he uploads it to the ALFRED marketplace in order to start the review process. As soon as the review process has been finished, Dave will be notified about the state of his app – whether it has passed the review or not. After Dave has created and uploaded his app and it has passed the review process, Olivia will be able to find it on the marketplace, install it onto her ALFRED device and check her health status whenever she wants to.	

5 Functional and Non-functional Requirements

This chapter gives the functional and the non-functional requirements that are the main results of task 2.3 “User Stories and Requirements Analysis”. These requirements are based on the user group data analysis that was carried out by the project partners.

All the ALFRED user stories have:

- Identification number (ID)
- Title
- Specific user group
- Related work task
- Related Use Case
- Priority number

5.1 Prioritization Process of the User Stories

The prioritization process of the user stories gathered during the user requirements phase was based on all the feedback acquired from the users and from the literature review and taking into consideration the context of the ALFRED project with the scope of the four different pillars. When the ALFRED user stories were prioritized, the description of work (DOW) of the project as well as the collected user data were considered. In addition, all the project partners contributed in this prioritization process. See Table 18 for the definition of the priority scale used for the user stories.

Table 18: User Story Priority Definition

Priority	Definition
1	Very high priority. Will be implemented.
2	High priority. Important for the ALFRED system; an implementation is planned.
3	Normal priority. Good additions for the ALFRED system; will be implemented if resources are available.
4	Low priority. Extension points for the ALRED system; will only be integrated if possible through synergies to other user stories.
5	Out of the project scope. Will not be implemented.

In the prioritization process six different factors had an impact of the final priority of each user story. These factors were in the following order:

1. **Direct Input:** During the different focus group session the input from actual end user were collected. The input was summarized and structured and were on the one hand the basis for the user stories as well as a weighting factor for the importance of the user stories.
2. **Scope:** Based on the Description of Work as well as the Strategy Consensus Document (D2.1) user stories were weighted on how good they fitted in the scope of the project.

3. **Literature:** The literature review provided an academic point of view of the user stories. This is a notably important weighing factor to allow “behind the scene” factor to be considered.
4. **Interest:** Naturally in an R&D project, partners have different interest of the outcome and the results of a project. For this reason, the interest of – especially the technical – partners were taken as a minor weighting factor.
5. **Ethical:** The weighting based on ethical issues was only used to raise any user story priority. This was used to ensure a minimum focus on ethical related issues to prevent fatal dead ends were a practical usage of the project results would be impossible
6. **Regulatory:** As with the ethical weighting, the regulatory weighting was used only to raise any user story priority. This was done to prevent fatal dead ends were a practical usage of the project results would be impossible.

Based on these six different factors several discussions were held. During the discussions a mutual agreement on the priorities for each user story was found by the following approach:

- For “Direct Input” the end-user partner provided (for the developer-user this was done by the technical partners) a priority from 1 to 5 (see table x)
- For the “Scope” a weighting with a “+”, “-“ was given or a user story was marked as “out of scope”, which resulted in a priority of 5 with no further adaption for the user story
- For “Literature” an adjustment for the priority could be given by the end user partners by weighting the user story with a “+” or “-“
- For “Interest” an adjustment were done. Every partner could provide a “+” to an user story to increase the priority, but if no partner was interested in a user story a “-“ was given
- For “Ethical” or “Regulatory” an adjustment could be provided by an “+” to increase the priority of a user story

Note: During each discussion for the priority of the user stories it was highlighted that every partner should not just provide a wish list, but has to stay focused in order to create a priority list, were on the one hand a reflection of the six factors are considered, while not losing sight of the limited efforts due to the the scope of the project on the other hand.